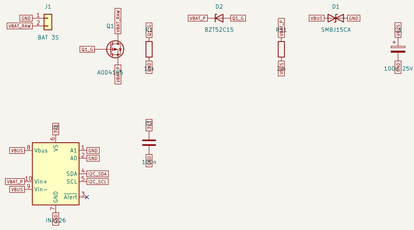
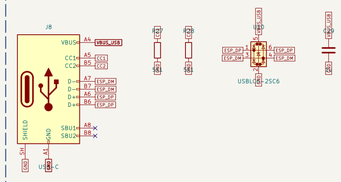
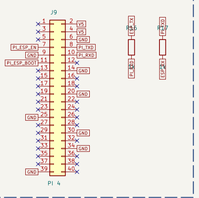
BATTERIJ 3S + STROOMMETING



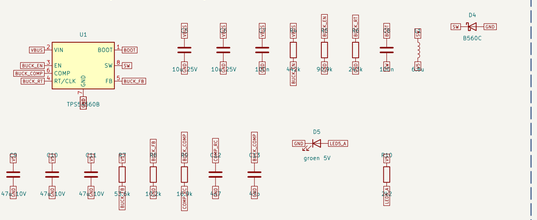
USB-C ESP32



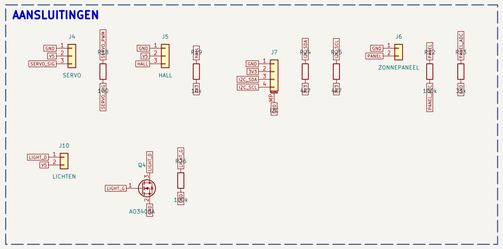
RASPBERRY PI 4 (40-PIN)



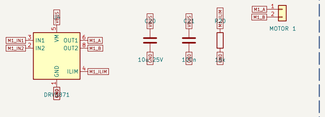
BUCK 5 V 5 A (PI 4 + SERVO)



AANSLUITINGEN



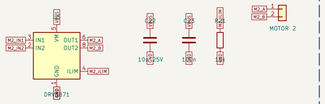
MOTOR 1 (DRV8871)



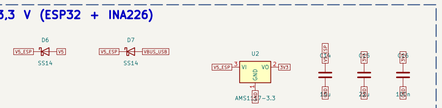
BEVESTIGING



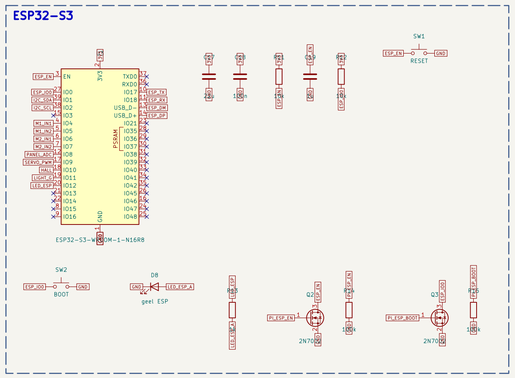
MOTOR 2 (DRV8871)



3.3 V (ESP32 + INA226)



ESP32-S3



SC-02 DRIVE BOARD v1.0 — ESP32-S3 + 2x DRV8871 + INA226 + 5 V 5 A buck, sits on the Raspberry Pi 4 like a HAT (generated by tools/build_sch.py from tools/design.py)
Battery 3S LiPo (9–12.6 V) -> P-MOSFET reverse protection -> 2 mOhm shunt (INA226 @ 0x40, I2C IO1/IO2) -> VBUS: motors + buck. Buck 5 V -> Pi 4 header pins 2/4, servo, hall, lights.
Motors: DRV8871 IN1/IN2 (M1: IO4/IO5, M2: IO6/IO7), current limit 64 / 18k = 3.55 A. Servo IO9 (100 R), hall A3144 IO10 (10k pull-up to 3V3), lights IO11 (AO3400A low side), panel ADC IO8 (100k/33k).
Pi link: ESP IO17 (TX) -> 1k -> Pi GPIO15 RXD (pin 10), Pi GPIO14 TXD (pin 8) -> 1k -> ESP IO18 (RX). Pi GPIO4 (pin 7) resets the ESP, Pi GPIO17 (pin 11) holds BOOT: flash it from the Pi with esptool.
The 40-pin socket sits on the BOTTOM side. Do not power the Pi from its own USB-C at the same time (the board feeds its 5 V pins directly).

ESP32-S3 + 2x DRV8871 + INA226, Pi 4 HAT
Kobe Van Robaeys
Sheet: /
File: sc02_drive.kicad_sch
Title: SC-02 drive board
Size: A1 | Date: 2026-10-18 | **Rev: 1.0**
KiCad E.D.A. 10.0.3 | Id: 1/1